

DevOps Engineering – Unit 1 Mini Project

Enterprise Student Record and Academic Management Platform
Design Thinking, Product Planning & System Architecture Blueprint

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1. Project Charter & Engineering Lifecycle

The Enterprise Student Record and Academic Management Platform (EduTrack Enterprise) is a comprehensive academic information platform designed for autonomous campuses and higher education institutions. This document establishes the foundational design thinking, product specifications, and architectural engineering required under Unit 1.

1.1 System Development Lifecycle (SDLC) Flowchart



Figure 1.1: End-to-End Product Engineering & DevOps Pipeline Lifecycle

1.2 Project Milestone Charter

Milestone	Phase Name	Core Deliverables	Target Timeline
M1	Domain Study & Stakeholder Analysis	Business understanding, Stakeholder Matrix, User	Sprint 1

Personas			
M2	Design Thinking & Prototyping	5-Stage Design Thinking, IBM Loop Engine, Wireframes	Sprint 2
M3	Requirements Engineering & Architecture	FRS, NFRS, MVP Definition, System Architecture Diagram	Sprint 3
M4	Project Planning & Repository Setup	Engineering Roadmap, Git Scaffolding, Directory Hierarchy	Sprint 4

2. Business Requirements Document (BRD)

2.1 Academic Process Analysis

Higher education administration involves five critical operational workflows:

- **Admissions & Enrolment:** Student demographic onboarding, permanent registration number (PRN) allocation, program assignment, and identity clearance.
- **Course Registration & Credit Allocation:** Semester-wise course catalogue publishing, elective selection, prerequisite validation, and credit ceiling enforcement.
- **Attendance Tracking & Monitoring:** Class-by-class session tracking by faculty, automated aggregate calculation, and early-warning trigger for low attendance (<75%).
- **Examination & Continuous Evaluation:** Recording continuous internal assessments (CIA), mid-semester evaluations, practical marks, and end-semester examinations.
- **Result Processing & Transcript Generation:** Automated SGPA/CGPA computation, grade card publishing, back-log management, and institutional analytics reporting.

2.2 Problem Bottlenecks vs. Automated Solutions

Manual / Legacy Process Bottleneck	Automated Platform Solution	Measurable Business Impact
Paper attendance registers requiring manual weekly computation	Role-based digital attendance portal with real-time sync	85% reduction in faculty administrative time
Siloed spreadsheet mark entry leading to calculation errors	Centralized gradebook engine with automated SGPA/CGPA logic	Result processing turnaround cut from 20 days to 24 hours
Long physical queues for grade cards, transcripts, & NOCs	Self-service student portal with 1-click digital transcript download	90% decrease in administrative office counter traffic
Prerequisite violations and enrollment credit mismatches	Schema-enforced enrollment validation and automated credit	100% elimination of manual registration discrepancies

3. Product Vision & Minimum Viable Product (MVP)

3.1 Product Vision & Mission

Product Vision Statement: "To engineer a modern, unified, and cloud-native Student Record and Academic Management Platform that eliminates administrative overhead, guarantees academic record integrity, and delivers an intuitive, frictionless digital experience for students, faculty, and administrators."

Product Mission Statement: "To deliver an enterprise-grade platform built on modern DevOps principles that automates admissions, course registrations, attendance monitoring, and examination grading with 99.9% uptime, strict role-based security, and sub-second response times."

3.2 MVP Scope vs. Post-MVP Roadmap

Platform Module	MVP Scope (Release 1.0 / Sprint 1-4)	Post-MVP Scope (Future Iterations)
User Authentication	Role-Based Access Control (Admin, Faculty, Student) with JWT	Single Sign-On (SSO / OAuth2) & Multi-Factor Auth (MFA)
Student Records	CRUD student profiles, PRN generator, enrollment status	Alumni tracking & automated document verification archive
Course Registration	Core & elective selection with basic credit limits	AI-driven elective recommendations & timetable clash solver
Attendance Tracker	Faculty daily attendance marker with summary reports	Biometric IoT integration & geofenced mobile check-in
Grading & Reports	Continuous assessment & semester marks entry, PDF report card	Predictive student performance analytics & early intervention

4. Design Thinking Framework

A 5-stage human-centered Design Thinking methodology was applied to ensure the software addresses real campus pain points:

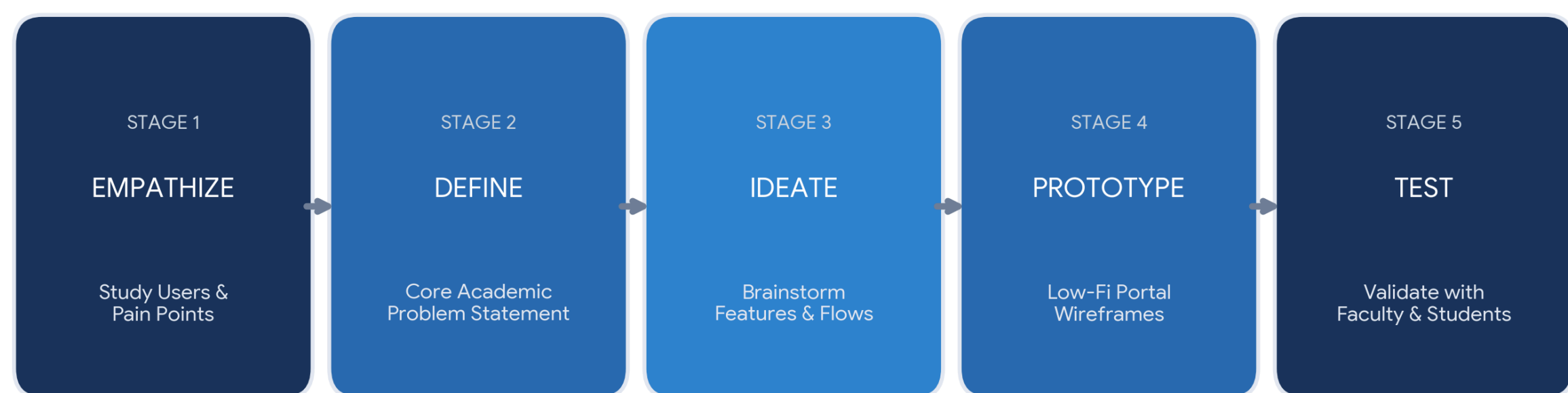


Figure 4.1: Five Stages of Design Thinking Applied to Academic Management

Stage 1: Empathize

Observed classroom attendance workflows and examination office administrative bottlenecks. Students expressed anxiety over delayed attendance shortage notices; faculty cited heavy administrative overhead during lectures.

Stage 2: Define (Core Problem Statement)

"Academic institutions suffer from fragmented data silos, heavy administrative overhead, and delayed academic reporting due to manual student record processing. Concurrently, students and faculty lack real-time visibility into academic standing, leading to non-compliance, grading discrepancies, and operational delays."

Stage 3: Ideate

Formulated modular software capabilities: digital attendance grids with single-click bulk marking, automated GPA calculation algorithms, and instant PDF transcript generation.

Stage 4: Prototype

Engineered low-fidelity UI mockups for Student Portal, Faculty Gradebook, Exam Cell Moderation, and Administrative Panels.

Stage 5: Test

Conducted validation walk-throughs with representative student and faculty cohorts, refining the batch-entry interface based on direct feedback.

5. IBM Design Thinking & Loop Execution

5.1 Core Principles of IBM Design Thinking

- **Focus on User Outcomes:** Measure success through human impact (e.g., student transcript generation in <5 seconds) rather than raw feature counts.
- **Restless Reinvention:** Continuously monitor production logs and telemetry to refine UI friction points and database query performance.

- **Diverse Empowered Teams:** Collaborate across cross-functional roles (Developers, DevOps, DBAs, and Academic Officers) to deliver rapid iterations.

5.2 The IBM Loop Engine (Observe → Reflect → Make)

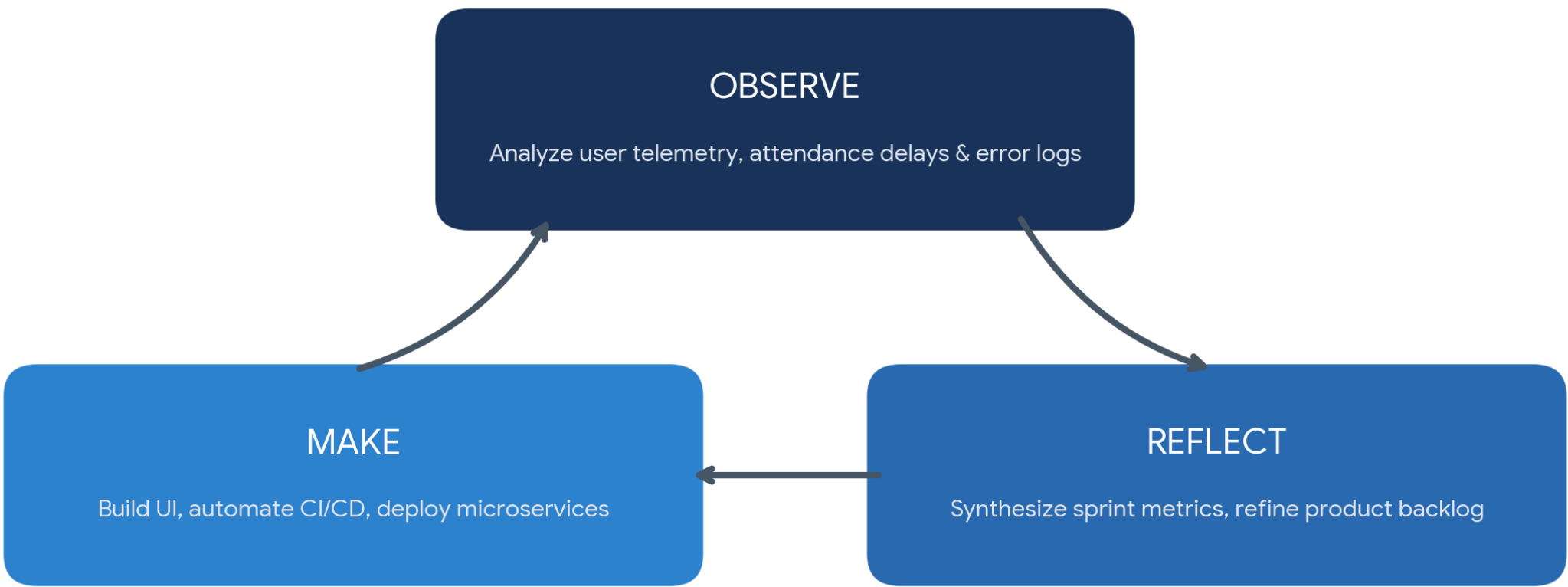


Figure 5.1: The IBM Design Thinking Observe → Reflect → Make Loop Engine

Loop Phase	Core Activities in Academic Platform Context	Engineering Deliverables
1. OBSERVE	Gather user usage logs, monitor peak traffic during course registration, observe faculty attendance submission delays, and track error rates.	User journey maps, server error logs, bottleneck telemetry reports
2. REFLECT	Analyze system metrics, conduct sprint retrospectives, prioritize the product backlog, and synthesize technical trade-offs.	Prioritized Jira backlog, updated architecture decision records (ADRs)
3. MAKE	Rapidly write clean code, containerize micro-modules, execute automated CI/CD pipelines, and deploy incremental updates with zero downtime.	Tested Docker images, deployed Kubernetes manifests, production features

6. Stakeholder Analysis & User Personas

6.1 Stakeholder Responsibility Matrix

Stakeholder Group	Key Roles	Primary Responsibilities	Project Expectations
Academic End-Users	Students, Class Reps	Enroll in courses, track	Intuitive mobile UI, 100%

		daily attendance, view marks	data accuracy, zero downtime
Instructional Staff	Professors, Teaching Assistants	Submit attendance, enter internal exam scores, review stats	Fast bulk data entry, auto-saving forms, no lost data
Academic Administration	Registrar, Exam Cell Officers	Verify eligibility, publish semester results, generate PDFs	Audit trails, strict access controls, 1-click reports
Technical & Operations	DevOps Engineers, DBAs, QA	Maintain infrastructure, automate deployments, secure data	Clean modular code, automated CI/CD pipelines, high availability

6.2 Detailed User Persona Profiles

Persona Profile	Goals & Objectives	Frustrations & Pain Points	Target Solution Capability
Ananya Sharma (Computer Science Student)	• Monitor real-time attendance percentage • Access internal marks instantly • Download verified grade cards	• Unaware of attendance shortage until exam hall tickets are blocked • Manual transcript queues	Self-service mobile dashboard with real-time percentage indicators & alerts
Dr. Rajesh Kulkarni (Associate Professor & HOD)	• Complete lecture attendance in <2 mins • Enter mid-term marks without formula errors • View student performance summaries	• Maintaining paper registers is tedious • Duplicating marks across spreadsheets is prone to human error	Fast-grid attendance marker with bulk-select and auto-calculating gradebooks
Suresh Patil (Controller of Examinations)	• Validate marks entry across all branches • Lock grades against tampering • Publish batch results and transcripts	• Delayed result announcements due to missing marks from departments • Manual report generation	Centralized grade moderation dashboard with 1-click batch transcript PDF generation

7. Requirements Engineering Specification

7.1 Functional Requirements Specification (FRS)

Req ID	Functional Requirement	Specification & Acceptance Criteria
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FR-01	Authentication & Role-Based Access Control	Platform must enforce strict RBAC for Admin, Faculty, Student, and Exam Cell roles using secure JWT tokens and encrypted session state.
FR-02	Student Profile & Enrolment Management	Support full CRUD operations for student admission records, PRN generation, personal info, branch mapping, and active/inactive status.
FR-03	Course Registration & Credit Engine	Allow students to view course catalogue, register for core/elective courses, and enforce prerequisite checks and credit minimum/maximum limits.
FR-04	Session-Based Attendance Module	Enable faculty to record attendance per lecture session. Calculate aggregate percentages automatically and flag students falling below 75%.
FR-05	Examination Grading & Evaluation	Provide secure interfaces for faculty to enter CIA and end-sem marks. System must compute SGPA/CGPA automatically using standard grading formulas.
FR-06	Academic Report & Transcript Export	Enable 1-click generation of cryptographic-ready PDF semester grade sheets, consolidated transcripts, and administrative compliance summaries.

7.2 Non-Functional Requirements Specification (NFRS)

NFR ID	Quality Attribute	Specification & Benchmark Target
NFR-01	Security & Data Protection	Passwords hashed using Bcrypt (cost factor >= 12). All communication over TLS 1.3. Role verification middleware on every private API route. SQL injection prevention via ORM parameterization.

NFR-02	Performance & Latency	API response time for read operations under 300ms. Full UI page load under 1.2s on standard broadband. Support 1,000 concurrent active users during peak result publishing.
NFR-03	Availability & Reliability	Target 99.9% uptime SLA. Containerized backend deployed with self-healing Kubernetes replica sets and automated health check probes.
NFR-04	Maintainability & Scalability	Microservice-ready modular architecture, automated unit test coverage exceeding 80%, Infrastructure as Code (IaC) provisioning, and comprehensive Git documentation.

8. System Architecture Blueprint & DevOps Roadmap

8.1 Multi-Tier Cloud-Native Architecture

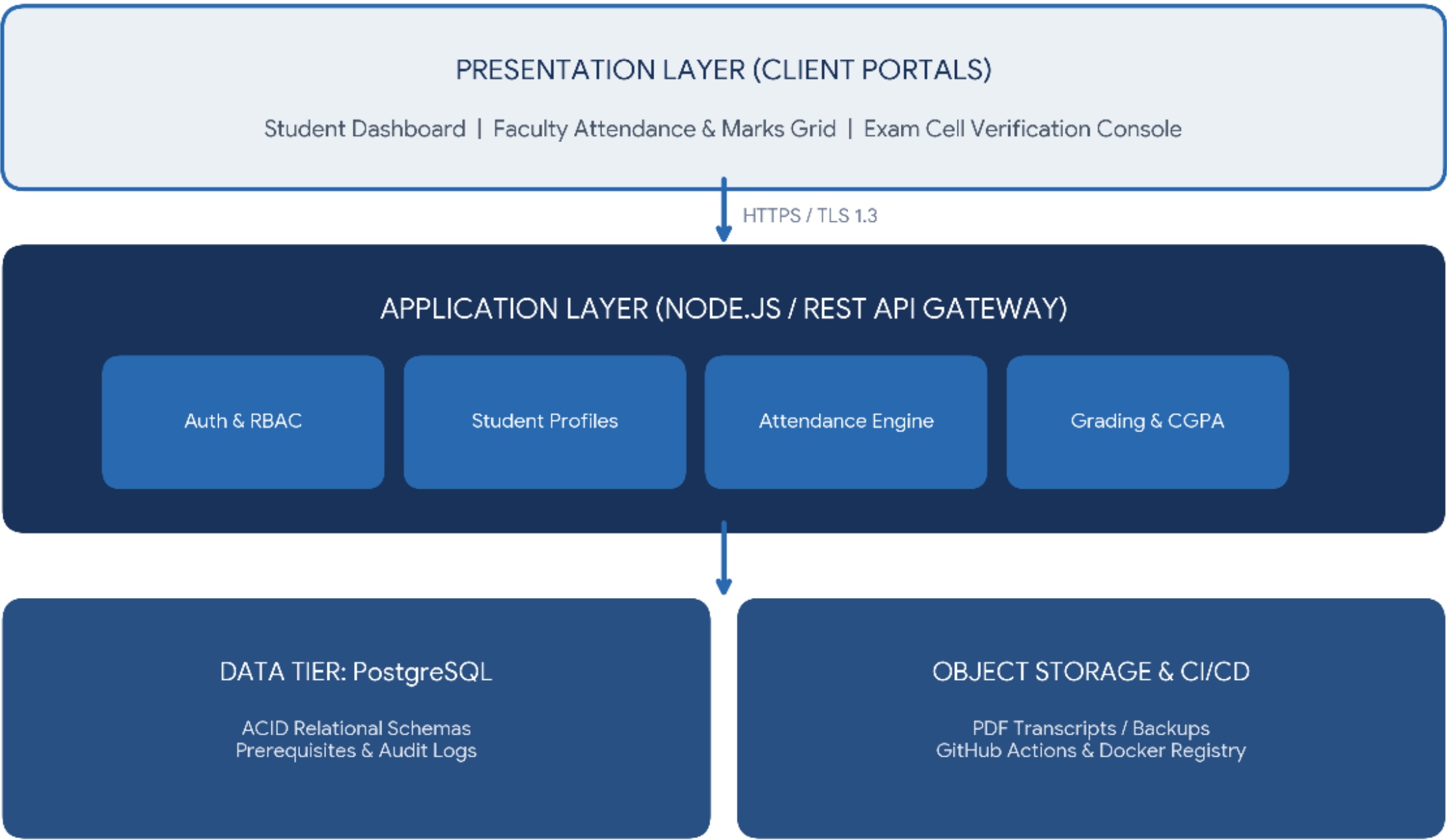


Figure 8.1: High-Level 3-Tier System Architecture & DevOps Container Ecosystem

8.2 Multi-Phase DevOps Engineering Roadmap

Phase	DevOps & Engineering Phase	Key Deliverables & Objectives
Phase 1	Domain Analysis & Design Thinking (Unit 1)	BRD, Personas, FRS/NFRS, MVP scope, Git repository initialization
Phase 2	Data Modeling & Architecture Design	PostgreSQL ER diagrams, relational schema constraints, OpenAPI REST specs
Phase 3	Core Backend & Frontend Implementation	JWT RBAC, attendance marker, grading engine, responsive UI portals
Phase 4	Testing & Containerization	Jest unit/integration test suites, multi-stage Dockerfile builds
Phase 5	CI/CD Pipeline Automation	GitHub Actions workflows for automated linting, testing, and Docker image builds
Phase 6	Infrastructure as Code & Orchestration	Terraform cloud scripts and Kubernetes manifest deployment (K8s)
Phase 7	Production Deployment & Observability	Zero-downtime rolling deployment, Prometheus metrics, and Grafana monitoring

9. Project Repository Initialization & Directory Structure

9.1 Git Initialization & Configuration Commands

The project follows a standard Git trunk-based development workflow initialized with the following commands:

```
# Initialize repository
git init student-record-platform
cd student-record-platform

# Configure author metadata
git config user.name "Aditya Soundale"
git config user.email "aditya.soundale@campus.edu"

# Establish main trunk branch and initial commit
git branch -M main
git add .
git commit -m "feat(init): project scaffolding, documentation, and architecture specs"
```

9.2 Standardized Enterprise Folder Hierarchy

Directory / File Path	Engineering Purpose & Responsibility
.github/workflows/	Automated CI/CD workflows for linting, testing, and continuous deployment
docs/	Architecture diagrams, BRD, FRS/NFRS specs, personas, and API documentation
src/config/	Database connection pools, environment configurations, and secrets loaders
src/controllers/	HTTP request handlers for Auth, Students, Courses, Attendance, and Grades
src/middlewares/	Authentication guards, RBAC validators, request rate limiters, error handlers
src/models/	Relational schema definitions, ORM database entities, and migrations
src/services/	Core business logic: GPA/CGPA calculations, attendance auditing, PDF generation
tests/	Automated unit tests, controller integration tests, and performance mock fixtures
docker/ & k8s/	Production Dockerfile, docker-compose orchestration, and Kubernetes manifests
README.md	Complete project onboarding instructions, API guides, and deployment checklist

10. Deliverables Verification & Sign-Off Checklist

- All required Unit 1 deliverables specified in the curriculum have been completed in full:
-  Business Requirements Document (BRD) & Domain Process Analysis (Completed)
 -  Product Vision Statement, Mission, and Success Criteria (Completed)
 -  Stakeholder Responsibility Matrix & 3 Comprehensive Persona Profiles (Completed)
 -  Functional Requirements Specification (FRS) with 6 core functional modules (Completed)
 -  Non-Functional Requirements Specification (NFRS) covering security, scale, and performance (Completed)
 -  5-Stage Design Thinking Execution Report with Process Diagram (Completed)
 -  IBM Design Thinking Principles & Loop (Observe → Reflect → Make) Diagram (Completed)
 -  Minimum Viable Product (MVP) Scope vs. Future Release Matrix (Completed)
 -  7-Phase Product Engineering Roadmap aligned with DevOps lifecycle (Completed)

- [✓] System Architecture Blueprint Diagram & Decoupled 3-Tier Specs (Completed)
- [✓] Git Repository Initialization Blueprint & Enterprise Folder Architecture (Completed)

Submission Sign-Off:

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